

ASCEND: PHYSICS
Philippine Physics Olympiad 2026 - Mock Exam
Test Questionnaire

Instructions:

- This is a 3-hour exam. However, you may opt to submit your Answer Sheet early.
- Calculators and Formula Sheets are prohibited.
- This Questionnaire consists of 10 pages, including this page, and the following parts:

	Page #'s
Instructions	1
Physical Constants and Mathematical Formulas	2
Part A: Multiple Choice (21 items, 42 points)	3 to 6
Part B: Fill-in-the-Blanks Questions (8 items, 32 points)	7 to 9
Part C: Problems Requiring Full Solutions (3 items, 21 points)	10

- The entire exam is worth 95 points. The questions for each part are equally weighted but are not necessarily equally difficult.
- You may keep the Questionnaire after the exam.
- No scratch paper will be provided and no paper other than that provided should be used. Use the white spaces and the blank pages of this questionnaire for your rough work.
- The Answer Sheet is provided separately. Ensure that your Answer Sheet remains clean and any mistakes in your answers or solutions are crossed out neatly.
- For problems where the answer is not a numerical value but an equation or expression, your answer may be expressed in terms of the parameters given in the problem (and the constant g if applicable). For example: $v = \sqrt{8gh}$.
- If you're answering this exam, it means you're one of the best students in school when it comes to physics. Good luck!

Possibly Useful Information

Physical Data and Constants

Quantity	Symbol	Value
Speed of Light	c	$3.00 \times 10^8 \text{ m/s}$
Planck's Constant	h	$4.14 \times 10^{-15} \text{ eV} \cdot \text{s} = 6.63 \times 10^{-34} \text{ J} \cdot \text{s}$
Radius of the Earth	R_E	6400 km
Gravitational Constant	G	$6.67430 \times 10^{-11} \text{ N} \cdot \text{m}^2 \cdot \text{kg}^{-2}$
Molar Gas Constant	R	$8.31446261815324 \text{ JK}^{-1}\text{mol}^{-1}$
Elementary Charge	e	$1.60217663 \times 10^{-19} \text{ C}$

Mathematical Information

$(1 + z)^p \approx 1 + pz + \frac{p(p - 1)}{2} z^2$ when $|pz| \ll 1$

$\cos(z) \approx 1 - \frac{1}{2} z^2$ when $|z| \ll 1$, and z is in radians

$\sin(z) \approx z - \frac{1}{6} z^3$ when $|z| \ll 1$, and z is in radians

Integration Formulas:

$\int \frac{dx}{x} = \ln |x| + C$

$\int \frac{x^2}{\sqrt{x + a}} dx = \frac{2}{15} \sqrt{x + a} (3x^2 - 4ax + 8a^2) + C$

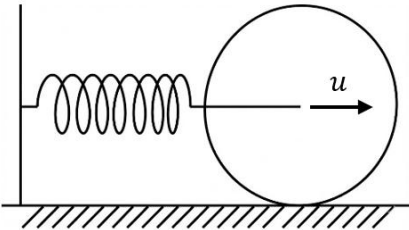
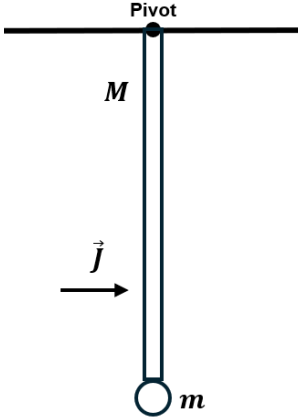
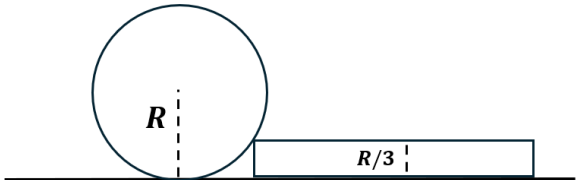
Differential Equations:

$z(x) = C_1 \sin(\sqrt{A}x) + C_2 \cos(\sqrt{A}x)$ where C_1 and C_2 are constants.

The general solution of $\frac{d^2z}{dx^2} - Az = 0$, where A is a positive constant, can be written as:

$z(x) = K_1 e^{\sqrt{A}x} + K_2 e^{-\sqrt{A}x}$ where K_1 and K_2 are constants.

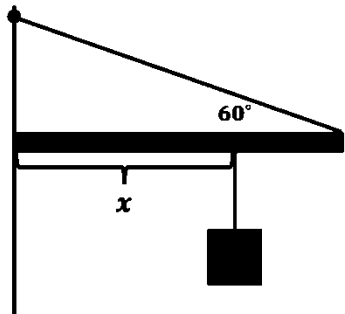
Philippine Physics Olympiad 2026 Mock Exam
Part A (Multiple-Choice, 21 items @ 2 points per item = 42 points)

A1. The escape speed from a uniform spherical planet of radius R is v_{esc}. A projectile is launched vertically from the surface of the planet with an initial speed $v_0 = \frac{1}{2} v_{esc}$. Neglecting air resistance and the effects of the planet's rotation, the maximum altitude reached by the projectile is: A) $R/4$ B) $R/3$ C) $R/2$ D) R E) $4R/3$	
A2. A particle is launched from the origin ($x = 0, y = 0$) with velocity u at an angle 30° to the horizontal. It lands on a coordinate $x = R_A$ on level ground. A second particle is launched from the same origin with the same speed u but at an angle 60°, landing at $x = R_B$. If the time of flight for the first particle is t_A and for the second is t_B, the ratio t_B/t_A is closest to: A) 1.7 B) 1.5 C) 1.0 D) 0.6 E) 0.5	
A3. A hollow cylinder of mass m is attached to a horizontal spring of constant k on a flat surface. When released from an amplitude A, it rolls without slipping with center-of-mass speed u. Suppose that the spring is cut in half and the cylinder is filled with a frictionless fluid of mass m. If the system is released from the same amplitude, the new speed is closest to: A) $0.77u$ B) $0.82u$ C) $1.15u$ D) $1.41u$ E) $2.00u$	
A4. A uniform rigid rod of mass M and length L hangs vertically from a frictionless pivot at its top end. A small sphere of mass $m=M/3$ and negligible radius is attached to the very bottom end of the rod. The system is initially at rest. An impulse $\vec{J} = J\hat{i}$ is delivered to the rod at a distance x below the pivot. There exists a specific impact point $x = x_p$ (called the center of percussion) such that the horizontal reaction force at the pivot is zero during the impact. This center of percussion is: A) $2L/3$ B) $3L/4$ C) $4L/5$ D) $5L/6$ E) $5L/8$	
A5. A uniform sphere has a radius R and mass m, as shown below.  What minimum force is needed to roll the sphere over a rigid plank of height $R/3$? A) $\frac{\sqrt{5}}{6}mg$ B) $\frac{\sqrt{5}}{3}mg$ C) $\frac{\sqrt{3}}{4}mg$ D) $\frac{\sqrt{3}}{2}mg$ E) $\frac{3}{2}mg$	

A6. A spherical planet has uniform mass density ρ and radius R . A straight tunnel is bored through the planet, passing through the center. A particle is released from rest at the surface. If the radius of the planet were changed to $2R$ while keeping the density ρ constant, the time taken for the particle to reach the opposite surface of the planet changes by a factor of:

- A) 2.0 B) 1.4 C) 1.0 D) 0.7 E) 0.5

A7. One end of a uniform meter stick is placed against a vertical wall. The other end is held by a lightweight cord that makes an angle 60° with the stick. The coefficient of static friction between the end of the meter stick and the wall is $\mu_s = \sqrt{3}/2$. A block of the same weight as the stick is suspended from the stick at a distance x from the wall. The minimum value of x for which the stick will remain in equilibrium is closest to:



- A) 0.67 B) 0.75 C) 0.83 D) 0.88 E) 0.95

A8. A horizontal disk of radius R rotates about a vertical axis through its center with constant angular speed ω . A particle of mass m is placed at a distance $R/2$ from the center of the disk. The coefficients of kinetic and static friction between the particle and the disk are $\mu_k = 1$ and $\mu_s = 4/3$, respectively. To keep the particle stationary relative to the disk, the maximum angular speed must be equal to $\sqrt{Cg/R}$, where C is closest to:

- A) 0.38 B) 0.67 C) 1.00 D) 1.33 E) 2.67

A9. A spring of negligible mass hangs vertically. The unstretched length of the spring is L . A block of mass m is attached to the spring when it is unstretched and then released from rest. The spring does not obey Hooke's law – the restoring force exerted by the spring when extended by a distance x has magnitude kx^2 , where k is a positive constant. When the spring has stretched a distance L , the speed of the block is $\sqrt{gL}$. The maximum distance that the spring stretches is:

- A) $2L$ B) $1.5L$ C) $1.2L$ D) $\sqrt{2}L$ E) $\sqrt{3}L$

A10. A car travels with a constant speed u towards a large, vertical, perfectly reflecting wall. The car sounds its horn, emitting sound waves of frequency f_0 . The speed of sound in the air is v (where $v > u$). The driver observes a beat frequency Δf due to the interference between the sound received directly from the horn and the sound reflected from the wall. The beat frequency is given by:

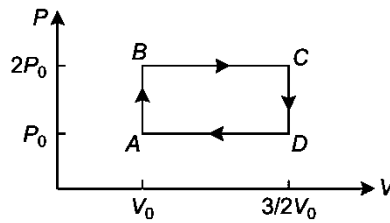
- A) $\frac{2u}{v-u}f_0$ B) $\frac{2u}{v+u}f_0$ C) $\frac{u}{v}f_0$ D) $\frac{2v}{v-u}f_0$ E) $\frac{u}{v-u}f_0$

A11. A layer of ice is forming on the surface of a pond where the air temperature is well below freezing. It takes a time t_0 for the ice layer to grow from a thickness of zero to a thickness d . How much additional time is required for the ice layer to grow from a thickness d to a thickness $2d$?

- A) t_0 B) $1.5t_0$ C) $2t_0$ D) $3t_0$ E) $4t_0$

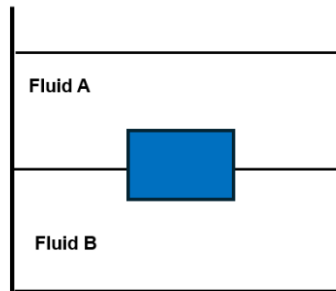
A12. A monatomic ideal gas is the working substance for a heat engine that undergoes the cyclic process ABCDA shown in the PV-diagram. The efficiency of this engine is closest to:

- A) 13% B) 20%
C) 33% D) 67% E) 71%



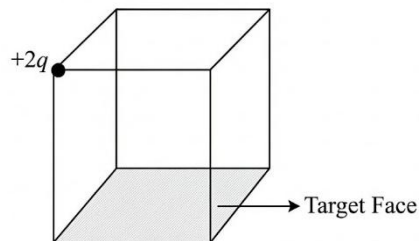
A13. A rectangular block of mass M and horizontal cross-sectional area A floats in equilibrium at the interface between two immiscible liquids of densities ρ_A and ρ_B (with $\rho_B > \rho_A$), as shown. The block is pushed down by a small vertical distance and released. The period of the resulting small oscillations is:

- A) $2\pi \sqrt{\frac{M}{gA(\rho_B + \rho_A)}}$ B) $2\pi \sqrt{\frac{M}{gA\rho_B}}$ C) $2\pi \sqrt{\frac{M}{gA(\rho_B - \rho_A)}}$
D) $2\pi \sqrt{\frac{M\rho_B}{gA(\rho_B - \rho_A)^2}}$ E) $2\pi \sqrt{\frac{M}{gA\rho_A}}$



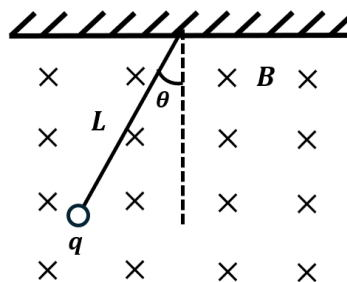
A14. A point particle of charge $+2q$ is placed exactly at one corner of a cube, as shown in the figure. The electric flux passing through one of the faces of the cube that is not touching the charge is equal to:

- A) $\frac{q}{3\epsilon_0}$ B) $\frac{q}{4\epsilon_0}$ C) $\frac{q}{6\epsilon_0}$ D) $\frac{q}{8\epsilon_0}$ E) 0



A15. A small sphere of mass m and charge $+q$ is suspended by a light, insulating thread of length L . The region is immersed in a uniform magnetic field of magnitude B directed into the page. The sphere is released from rest at an angle θ from the vertical. As the sphere swings back and forth, the maximum tension in the thread is:

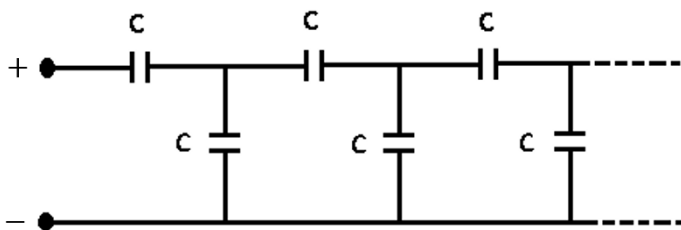
- A) $3mg + 3qB\sqrt{2gL}$ B) $3mg - qB\sqrt{2gL}$
C) $3mg + qB\sqrt{2gL}$ D) $mg + qB\sqrt{2gL}$
E) $3mg$



A16. An electric dipole consists of two small spheres of mass m and charge $+q$ and $-q$, connected by a rigid, insulating rod of length $2L$ and mass $5m$. The dipole is placed in a uniform electric field of magnitude E . Initially aligned with the field, it is given a small angular displacement and released. The angular frequency of oscillation of the dipole is $\sqrt{\alpha \frac{qE}{mL}}$, where α is closest to:

- A) 0.55 B) 0.82 C) 1.00 D) 1.20 E) 1.73

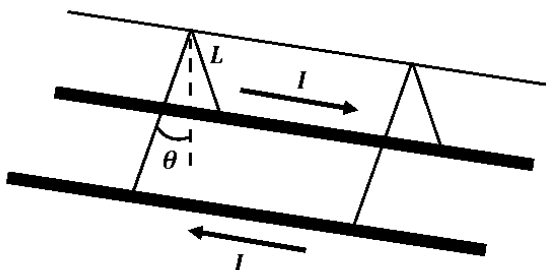
A17. A network of capacitors of capacitances C extends to infinity to the right as shown in the figure.



The equivalent capacitance of the infinite network is aC , where a is closest to:

- A) 0.50 B) 0.62 C) 1.00 D) 1.62 E) 2.62

A18. Two long, parallel wires hang by cords of length L from a common axis. The wires have a mass per unit length λ and carry the same current I in opposite directions, as shown.



The current in each wire when the cords hang at a small angle θ with the vertical is:

- A) $\sqrt{\frac{2\pi\lambda L g \theta^2}{\mu_0}}$ B) $\sqrt{\frac{4\pi\lambda L g \theta^2}{\mu_0}}$ C) $\sqrt{\frac{2\pi\lambda L \theta}{\mu_0}}$
D) $\sqrt{\frac{4\pi\lambda L g \theta}{\mu_0}}$ E) $\sqrt{\frac{\pi\lambda L g \theta^2}{2\mu_0}}$

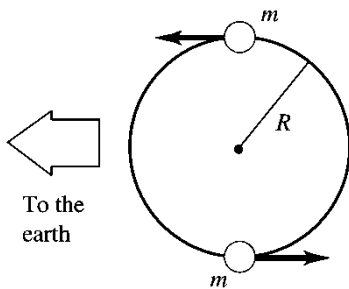
A19. A certain transparent material has a refractive index of $n = 2.0$. The critical angle for total internal reflection at the interface between this material and air ($n_{air} \approx 1.0$) is:

- A) 30° B) 42° C) 45° D) 60° E) 90°

A20. A meter stick is at rest in a laboratory frame and makes an angle of 30° with the x -axis. An observer in a spaceship moving with speed u along the x -axis measures the angle of the stick to be 60° with the x -axis. In terms of the speed of light c , the spaceship's speed u is:

- A) $c/2$ B) $\sqrt{2}c/3$ C) $3c/4$ D) $2c/3$ E) $2\sqrt{2}c/3$

A21. Two identical stars of mass m orbit their common center of mass in circular orbits with a period T . The system is viewed edge-on from Earth, as shown in the figure. The light from the stars shows a periodic Doppler shift, where the maximum fractional frequency shift from the rest frequency is $\delta = |\Delta f/f_0|$. Assuming $\delta \ll 1$, the separation distance between the two stars is:



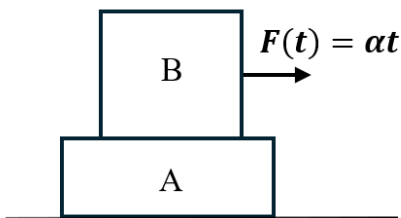
- A) $\frac{c\delta T}{2\pi}$ B) $\frac{c\delta T}{\pi}$ C) $\frac{2c\delta T}{\pi}$ D) $\frac{c\delta T}{4\pi}$ E) $\frac{c\delta T}{2}$

Philippine Physics Olympiad 2026 Mock Exam

Part B (Fill-in-the-Blanks, 8 items @ 4 points per item = 32 points)

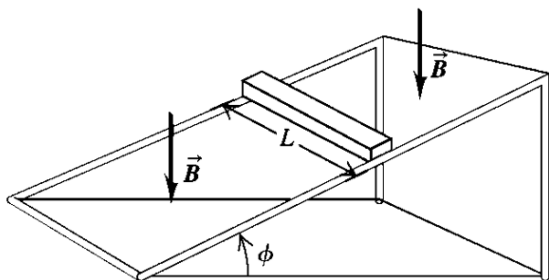
This section consists of eight (8) problems. Only final answers are needed. Write your final answers in the space provided in the Answer Booklet.

B1. Block A of mass m lies on a smooth horizontal surface. Block B of identical mass m is placed on top of Block A. The coefficient of friction between the blocks is $\mu_s = \mu_k = \mu$. The system is initially at rest. At time $t = 0$, a horizontal force growing linearly with time, $F(t) = \alpha t$ (where α is a positive constant), is applied to Block B as shown in the figure.



Let t_0 be the time when the bar begins to slide relative to the plank. Find the total heat generated by friction during the time from $t = t_0$ to $t = 2t_0$. Express your answer in terms of μ , m , g , and α .

B2. A metal bar with length L , mass m , and resistance R is placed on frictionless metal rails that are inclined at an angle $\phi = 60^\circ$ above the horizontal. The rails have negligible resistance. A uniform magnetic field of magnitude B is directed vertically downward, as shown in the figure.

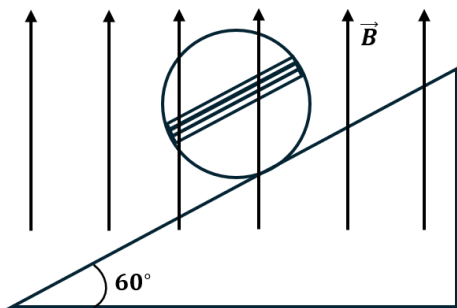


The bar is released from rest and slides down the rails. Determine the power dissipated in the resistor after the bar has reached its terminal velocity. Express your answer in terms of R , m , g , L , and B .

B3. A small rock of mass m is suspended from the ceiling of an elevator by a light cord. The rock is totally immersed in a bucket of water that sits on the floor of the elevator. The rock does not touch the sides or bottom of the bucket and is at a vertical distance H relative to the bucket. When the elevator is at rest, the tension in the cord is measured to be T_0 . The elevator now begins to accelerate upward with a constant acceleration $3g$. At the exact moment the elevator starts moving, the cord is cut. Determine the time t_{sink} it takes for the rock to sink to the bottom of the bucket. Express your answer in terms of H , g , m , and T_0 only.

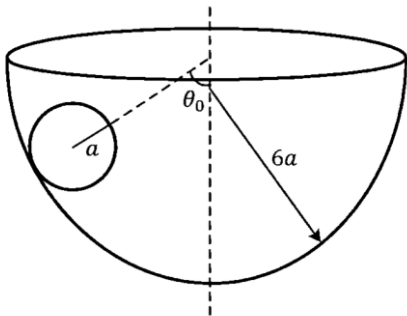
B4. A uniform steel chain of length L is placed on a rough horizontal table. The coefficient of kinetic friction between the chain and the table is $\mu_k = 1/2$, while the coefficient of static friction is $\mu_s = 3/4$. Initially, a length $L/2$ hangs vertically over the edge, and the chain is held at rest. The chain is then released and begins to slide off the table. The chain has a specific heat c and a coefficient of linear thermal expansion α_0 . Determine the fractional change in length of the chain at the instant the last link leaves the table. Express your answer in terms of α_0 , g , L , and c . Assume that the chain absorbs all the thermal energy generated by friction and that the heat distributes uniformly throughout the chain.

B5. A uniform non-conducting sphere has mass m and radius R . A flat compact coil of wire with N turns is wrapped tightly around it, with each turn concentric with the sphere. The sphere is placed on an inclined plane making an angle 60° with the horizontal, so that the coil is parallel to the inclined plane. A uniform magnetic field of magnitude B directed vertically upward exists in the region of the sphere, as shown in the figure.



What current in the coil will enable the sphere to rest in equilibrium on the inclined plane? Express your answer in terms of m , g , N , R , and B .

B6. A uniform hollow sphere of radius a rolls without slipping inside a fixed large spherical bowl of radius $6a$. The sphere is released from rest at a small angle θ_0 from the vertical, as shown below (angle in figure is exaggerated).



Determine time it takes for the center of mass of the sphere to first reach the bottom of the bowl. Express your answer in terms of a and g only.

B7. At $t = 0$, a concave mirror of focal length f is moving with a velocity $\vec{v}_m = -2u\hat{i}$ while a point object located on the principal axis is moving with a velocity $\vec{v} = u\hat{i}$. Determine the velocity of the image with respect to the ground at the instant the object is at a distance $2f$ from the mirror,. Express your answer in terms of u only.

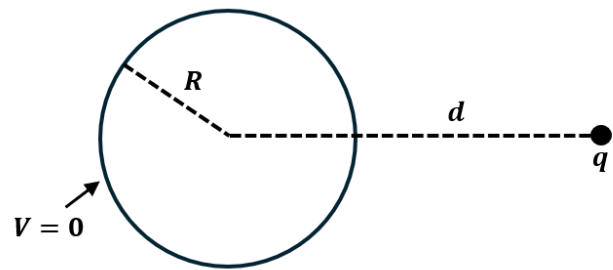
B8. A free particle of mass m is initially at rest. A photon with energy mc^2 collides head-on with the particle. After the collision, the photon scatters perfectly backward along its original path. Determine the speed of the free particle after the collision. Express your answer as a fraction of the speed of light c .

Philippine Physics Olympiad 2026 Mock Exam**Part C (Problems Requiring Full Solutions, 3 items @ 7 points per item = 21 points)**

This section consists of three (3) problems. Full solutions are required. Write your solution and final answers in the space provided in the **Answer Booklet**. Box your final answers.

C1. The Adiabatic Atmosphere. Consider a planetary atmosphere consisting of an ideal gas of molar mass μ . The atmosphere is in hydrostatic equilibrium, and the acceleration due to gravity g is assumed to be constant. The atmosphere is unstable to convection unless the temperature gradient satisfies the “adiabatic lapse rate”. Assuming the pressure P and density ρ are related by the adiabatic condition $P \propto \rho^\gamma$ where γ is the adiabatic index, express the magnitude of the vertical temperature gradient $\left| \frac{dT}{dy} \right|$ in terms of g , μ , γ , and the ideal gas constant R .

C2. Method of Images. A point charge $+q$ is held at a distance d from the center of a grounded conducting sphere of radius R , as shown in the figure. “Grounded” means that the potential of the sphere is fixed at zero ($V = 0$).

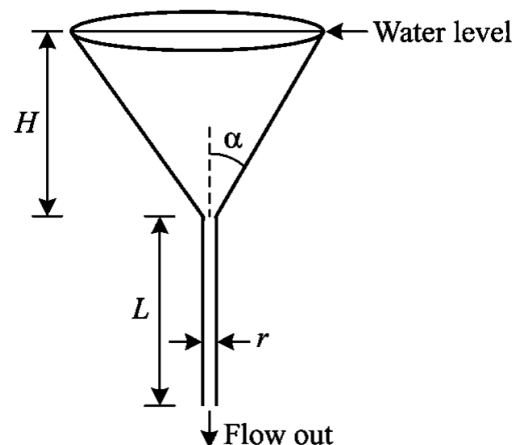


A) Calculate the magnitude of the electrostatic force of attraction F between the point charge $+q$ and the sphere in terms of q , R , d , and ϵ_0 . [4 pts]

B) If the charge $+q$ is released from rest at distance $d = 5R$, calculate the speed of the charge when it reaches a distance $d = 2R$. [3 pts]

Hint: This system can be modeled by replacing the sphere with a single “image charge” q' located at a specific distance b from the center of the sphere along the line connected the center and $+q$. Then the boundary conditions can be used to related q' and b to q , d , and R .

C3. The Clepsydra. An ancient water clock, a “clepsydra”, is a device designed to measure short intervals of time. It consists of a large conical vessel of height H and half-angle α , fitting with a long, thin, vertical discharge tube of length L and constant cross-sectional radius r . The entire device is initially filled with water, as shown in the figure. The water drains out from the bottom of the tube. Assume that the cross-sectional area of the water surface of the cone is always much larger than the cross-sectional area of the tube.



A) Determine the time t_{empty} required to empty the conical section, expressing your answer in terms of H , α , L , r , and g . [5 pts]

B) Consider a specific device where $H = 3L$. Furthermore, the dimensions are chosen such that the cross-sectional area of the cone at a height L above the junction is exactly 15 times the cross-sectional area of the discharge tube. In this case, t_{empty} can be expressed in the form $N\sqrt{L/2g}$ where N is an integer. Determine the value of N . [2 pts]